

American International University-Bangladesh (AIUB)

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Section: B

Software Quality Assurance and Testing

StayEase - Hotel Room Booking Management System

A Report submitted
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Software Test Plan

for

<StayEase>

Version 1.0 approved

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Table of Contents

Revision History 3
1. TEST PLAN IDENTIFIER: StayEase-VER1.0 4
2. REFERENCES 4
3. INTRODUCTION 4
3.1 Background to the Problem 4
3.2 Solution to the Problem 4
4. REQUEIREMNT SPECIFICATION 6
4.1 System Features 6
4.2 System Quality Attributes 8
4.3 System Interface 10
4.4 Project Requirements 23
5. FEATURES NOT TO BE TESTED 26
6. TESTING APPROACH 26
6.1 Testing Levels 26
6.2 Test Tools 27
6.3 Meetings 27
7. TEST CASES/TEST ITEMS 29
8. ITEM PASS/FAIL CRITERIA 60
9. TEST DELIVERABLES 61
10. STAFFING AND TRAINING NEEDS 62
11. RESPONSIBILITIES 63
12. TESTING SCHEDULE 63
13. PLANNING RISKS AND CONTINGENCIES 64

Revision History

Revision	Date	Updated by	Update Comments

1. TEST PLAN IDENTIFIER: StayEase-VER1.0

2. REFERENCES

I. Software Requirement Specification (SRS) of StayEase
II. Course Material: Software Quality Assurance and Testing

3. INTRODUCTION

3.1 Background to the Problem

Hotel room booking in Bangladesh — for all kinds of hotels including small and mid-size hotels, guest houses, and private room hosts — remains heavily dependent on phone calls, walk-in visits, and informal, paper-based arrangements. Unlike large international hotel chains that have invested in property management systems, the vast majority of small and mid-size accommodation providers in the country still rely on manual processes to track room availability, record bookings, and manage guest communication. The root cause of this problem lies in the absence of a centralized, real-time availability-management mechanism accessible to these smaller hosts.

This problem is important to consider for several reasons:

- **Guest-side impact:** Travelers have little to no visibility into real-time room availability, pricing, or amenities across many small hosts, forcing them to rely on word-of-mouth or repeated phone calls, which is time-consuming and unreliable, especially for guests booking from another city or unfamiliar with the local market.
- **Host-side impact:** Small and mid-size hotel owners lose potential revenue from missed bookings, double-booking penalties, and poor occupancy planning.
- **Market-side impact:** International platforms such as Booking.com do serve part of the Bangladeshi market, but they charge high commissions (often significantly eating into a small host's margin) and are not specifically designed around the pricing sensitivity, payment habits (e.g., mobile financial services like bKash/Nagad), and property scale typical of the local hospitality sector. This leaves a significant gap: a large segment of small-to-mid-size Bangladeshi hosts remain effectively excluded from a fair, low-commission, digitally-enabled booking ecosystem.

3.2 Solution to the Problem

To address this gap, we propose StayEase, a web-based Hotel Room Booking Management System designed specifically for the local market of Bangladesh. StayEase provides a unified, commission-transparent platform where verified room hosts can list their properties and guests can discover, compare, and book rooms entirely online, with real-time visibility.

Why this solution is appropriate: The core weakness identified in the problem background is the lack of a shared, real-time source for room availability. StayEase directly solves this by maintaining a centralized availability calendar per listing that is updated instantly the moment a booking is confirmed and paid for, eliminating the possibility of two guests being confirmed for the same room on overlapping dates. By building the platform around local payment gateways (SSL Commerz and bKash) and a lower, transparent service fee rather than the high commissions charged by global platforms, the solution is tailored to the economics and payment habits of the Bangladeshi hospitality market rather than being an imported model.

Feasibility relative to business objectives: The proposed solution is technically and commercially feasible against the stated business objectives:

1. Real-time availability locking upon payment confirmation directly satisfies the objective of eliminating double-booking conflicts.
2. A responsive, mobile-friendly design (given that most guests in Bangladesh will access the platform via smartphone) satisfies the objective of a mobile-friendly guest experience.
3. A dedicated Admin subsystem for listing approval and dispute resolution
4. A small, fixed service fee applied per completed booking (rather than a high blanket commission) satisfies the objective of generating sustainable platform revenue while remaining attractive to cost-sensitive hosts.

Purpose and benefits of the software: StayEase is a three-sided web platform serving Guests, Hotel Room Hosts, and Administrators. Its purpose is to digitize the hotel room reservation process in Bangladesh end-to-end — from search and discovery, through booking and secure payment, to post-stay review — while giving hosts independent control over their listings, pricing, and availability without relying on third-party agents. Key benefits include reduced booking friction for guests, improved occupancy and revenue predictability for hosts, lower commission costs compared to global alternatives. The overarching goal is to improve occupancy rates for small-to-mid-size properties across Bangladesh while building guest confidence in a transparent, locally-relevant booking experience.

Existing studies and available solutions: Global platforms such as **Booking.com** and **Airbnb** represent the closest existing solutions in this problem space. Booking.com offers a mature, high-volume room search and booking workflow, and Airbnb offers strong host-facing listing and availability-calendar management features. However, neither platform is purpose-built for the Bangladeshi market: their commission structures are high relative to local room prices, their payment integrations are not centered on locally dominant mobile financial services, and their host-verification and support processes are not tailored to the scale of small, independent Bangladeshi hoteliers. At present, there is no dominant, commission-transparent platform built specifically for this market segment — which is the specific gap StayEase is designed to fill.

4. REQUEIREMNT SPECIFICATION

4.1 System Features

4.1.1 User Authentication & LogIn

1.1 The system shall allow Users, Host, and Admins to register with a unique email address and a password of at least 8 characters.
1.2 The system shall send a verification email upon registration; the account remains inactive until the user verifies their email.
1.3 The system shall allow registered users to log in using their email address and password.
1.4 The system shall verify login credentials against the database; if correct, it shall redirect the user to their respective home dashboard based on their role.
1.5 If the username or password entered is incorrect, the system shall display an error message and generate a one-time verification code sent to the registered email for the user to retry login.
1.6 If a user exceeds 3 consecutive failed login attempts, the system shall temporarily block the account login for one hour. *[Optional]*
1.7 The system shall provide a 'Forgot Password' option that sends a time-limited reset link to the registered email address.
1.8 The system shall enforce role-based access control (RBAC), ensuring each user class can only access features relevant to their role.
1.9 The system shall automatically invalidate the user session after 30 minutes of inactivity.

Priority Level: High
Precondition: User has a valid registered email address and is connected to the internet.

4.1.2 Room Search and Discovery

2.1 The system shall allow users to search for rooms by entering a location, check-in date, check-out date, and number of guests.

2.2 The system shall display only rooms with confirmed availability for the selected date range in search results.

2.3 The system shall allow users to filter search results by price range, room type (Single, Double, Suite, Family), amenities, and star rating.

2.4 The system shall allow users to sort search results by price (low to high / high to low) and by guest rating.

2.5 The system shall display each room listing with at least one photo, a title, location, price per night, and average star rating.

2.6 The system shall display a map view showing the geographic locations of search results based on the entered location.

Priority Level: High
Precondition: User is browsing as a guest or is logged in as a registered User.

4.1.3 Room Booking and Payment

3.1 The system shall allow a logged-in User to initiate a booking for an available room by selecting check-in and check-out dates and confirming guest count.
3.2 The system shall calculate and display the total booking cost before payment confirmation, including the nightly rate, service fee, and applicable tax.
3.3 The system shall allow users to complete payment via the integrated SSL Commerz or bKash payment gateway.
3.4 Upon successful payment, the system shall mark the room as unavailable for the booked dates and immediately update the availability calendar.
3.5 The system shall generate a unique Booking ID and send a booking confirmation to the user's registered email upon successful payment.
3.6 The system shall allow users to cancel a booking in accordance with the host's defined cancellation policy (Flexible, Moderate, or Strict).
3.7 The system shall maintain a complete booking history accessible by the user in their profile dashboard.

Priority Level: High
Precondition: User is logged in and has selected an available room with valid check-in and check-out dates.

4.1.4 Room Listing Management (Hotel Room Host)

4.1 The system shall allow a verified Host to create a new room listing by providing a title, description, location, room type, amenities, pricing, and photos.

4.2 The system shall allow Host to upload up to 10 images per listing, each not exceeding 5 MB in size (JPG/PNG only).

4.3 The system shall allow Host to set a standard nightly price as well as optional weekend and seasonal pricing.

4.4 The system shall provide Host with an availability calendar to open or block specific dates for their listings.

4.5 The system shall notify Host via email and SMS when a new booking is made or cancelled for their listing.

4.6 The system shall allow Host to deactivate or delete a listing.

4.7 The system shall provide Host with a revenue dashboard displaying total earnings, booking count, and monthly occupancy rate.

Priority Level: High
Precondition: Host account is registered, email-verified, and approved by the Admin.

4.1.5 Admin Dashboard and Platform Management

5.1 The system shall provide Admins with a dashboard displaying key platform metrics: total users, active listings, total bookings, and platform revenue.

5.2 The system shall allow Admins to view, approve, suspend, or permanently ban any User or Host account.

5.3 The system shall require Admin approval before a newly submitted Host listing becomes publicly visible.

5.4 The system shall allow Admins to remove any listing that violates platform policies.

5.5 The system shall allow Admins to review and resolve booking disputes reported by Users or Host.

5.6 The system shall allow Admins to generate reports on user activity, revenue, and booking trends, exportable as PDF or Excel.

Priority Level: High
Precondition: Admin is authenticated with admin-level credentials.

4.1.6 Review and Rating System

6.1 The system shall allow a User to submit a star rating (1–5) and a written review for a room only after their check-out date has passed.

6.2 The system shall display the average star rating and all approved guest reviews on each room's detail page.

6.3 The system shall allow users and hosts to report a review to the admin for moderation if it contains inappropriate content.

Priority Level: Medium
Precondition: User has a completed (checked-out) booking for the room being reviewed.

4.1.7 Notification System

7.1 The system shall send automated email notifications for booking confirmation, booking cancellation, payment receipt, and a check-in reminder 24 hours before arrival.

7.2 The system shall send an SMS alert to the Host when a new booking is made or cancelled on their listing.

Priority Level: Medium
Precondition: User has a valid registered email address and/or phone number.

4.2 System Quality Attributes

Security: The system shall implement robust security measures, including data encryption, access control, and authentication mechanisms, to protect sensitive user information and prevent unauthorized access or data breaches.

Priority Level: High

Performance: Any page in the system shall fully load within 3 seconds under normal operating load of up to 2,500 concurrent users. Room search results shall be returned within 2 seconds. The system shall provide a satisfactory user experience without measurable performance degradation.

Priority Level: High

Usability: The system interface shall be intuitive and user-friendly, allowing users to navigate through the application and perform tasks with minimal training or guidance. A first-time user with no prior training shall be able to complete a room booking within 5 minutes of account registration

Priority Level: High

Reliability: The system shall maintain 99.5% uptime, ensuring that it is consistently available for users to access and perform necessary tasks. All booking and payment transactions shall be atomic — if any step fails, the entire transaction shall be rolled back to prevent partial bookings or double charges.

Priority Level: High

Maintainability: The system shall be designed with modular and well-documented code, facilitating ease of maintenance, updates, and enhancements by system administrators and developers. New features shall be deployable via a zero-downtime deployment strategy.

Priority Level: Medium

Scalability: The system architecture shall support horizontal scaling by adding additional server nodes to handle traffic growth beyond the initial 5,000-user target. All static assets including images, CSS, and JavaScript shall be served through a Content Delivery Network (CDN) to reduce server load.

Priority Level: Medium

Interoperability: The system shall be compatible with various operating systems, web browsers, and devices, ensuring seamless integration and accessibility for users across different platforms.

Priority Level: Medium

4.3 System Interface

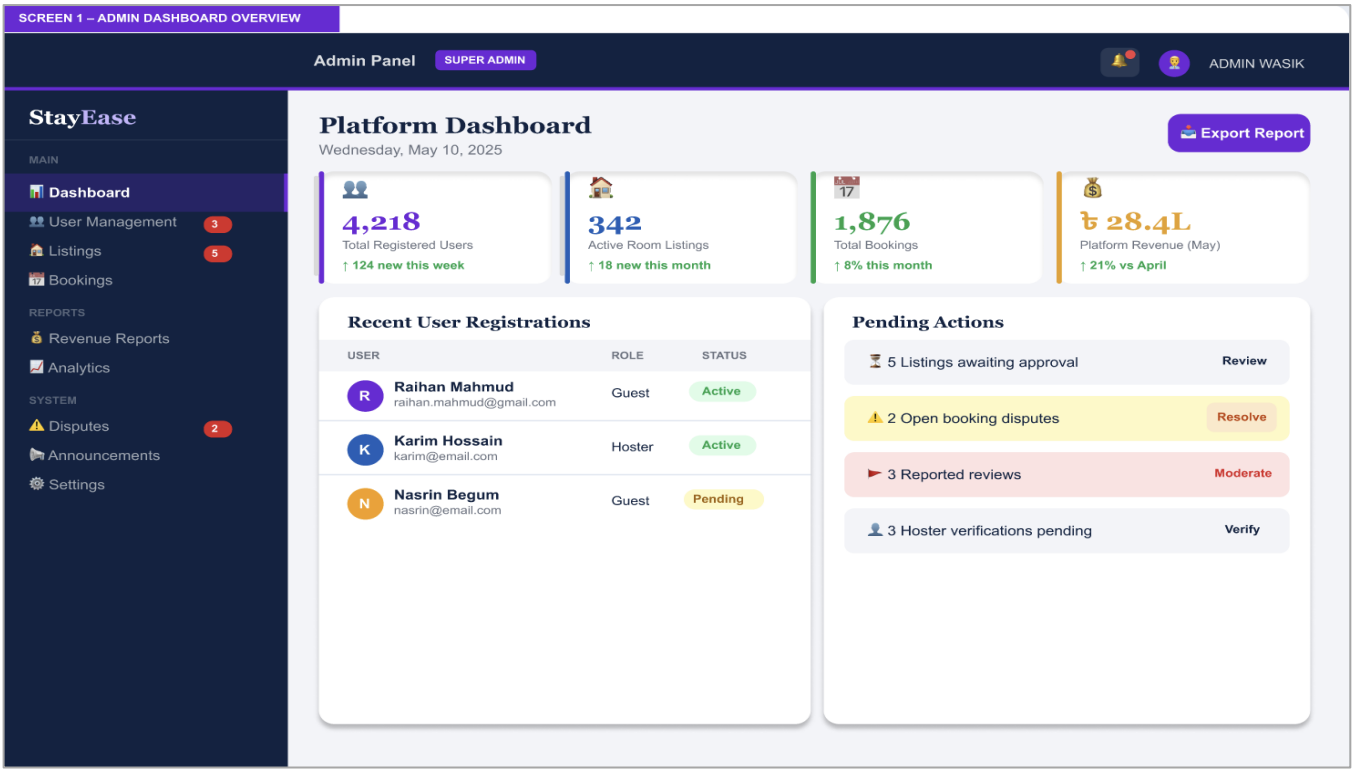


Fig 4.3.1: Admin Dashboard Overview

Description: This screen is the landing page for the Admin role, giving a snapshot of the entire StayEase platform. Four summary cards display total registered users, active room listings, total bookings, and platform revenue for the month, each with a week-over-week or month-over-month growth indicator. The 'Recent User Registrations' panel lists newly signed-up Guests and Hosts along with their verification status, while the 'Pending Actions' panel surfaces items requiring admin attention — pending listing approvals, open booking disputes, reported reviews, and host verification requests. An 'Export Report' button in the top-right allows the admin to download platform metrics.

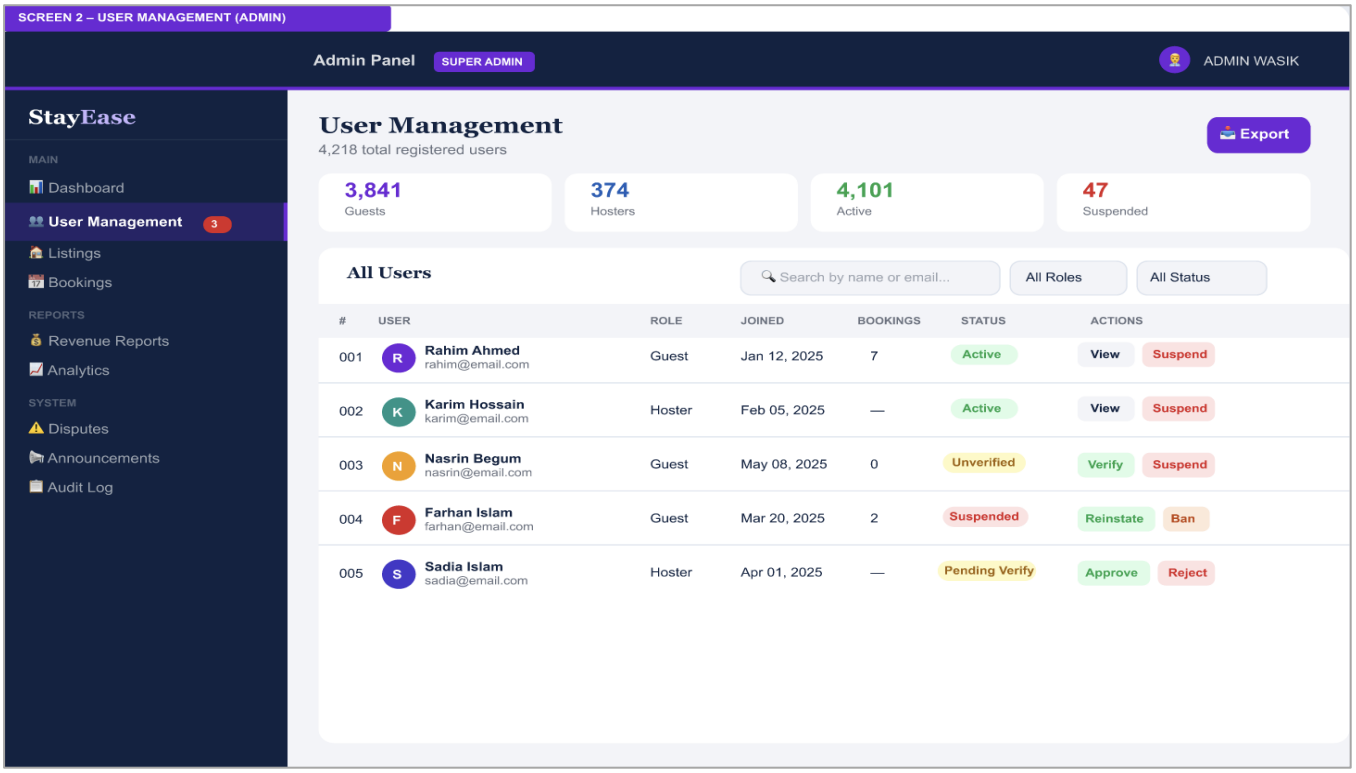


Fig 4.3.2: User Management (Admin)

Description: This screen allows the Admin to view and manage all registered users on the platform. Summary tiles at the top break down the user base by role (Guests, Hosts) and status (Active, Suspended). A searchable, filterable table lists each user with their role, join date, number of bookings, current status, and available actions. Depending on a user's current state, the Admin can View, Suspend, Ban, Verify, Approve, or Reject an account directly from this table. This screen implements FR 5.2 (Admin can view, approve, suspend, or ban any User or Host account) and supports the platform's governance and quality-control objectives described in the SRS.

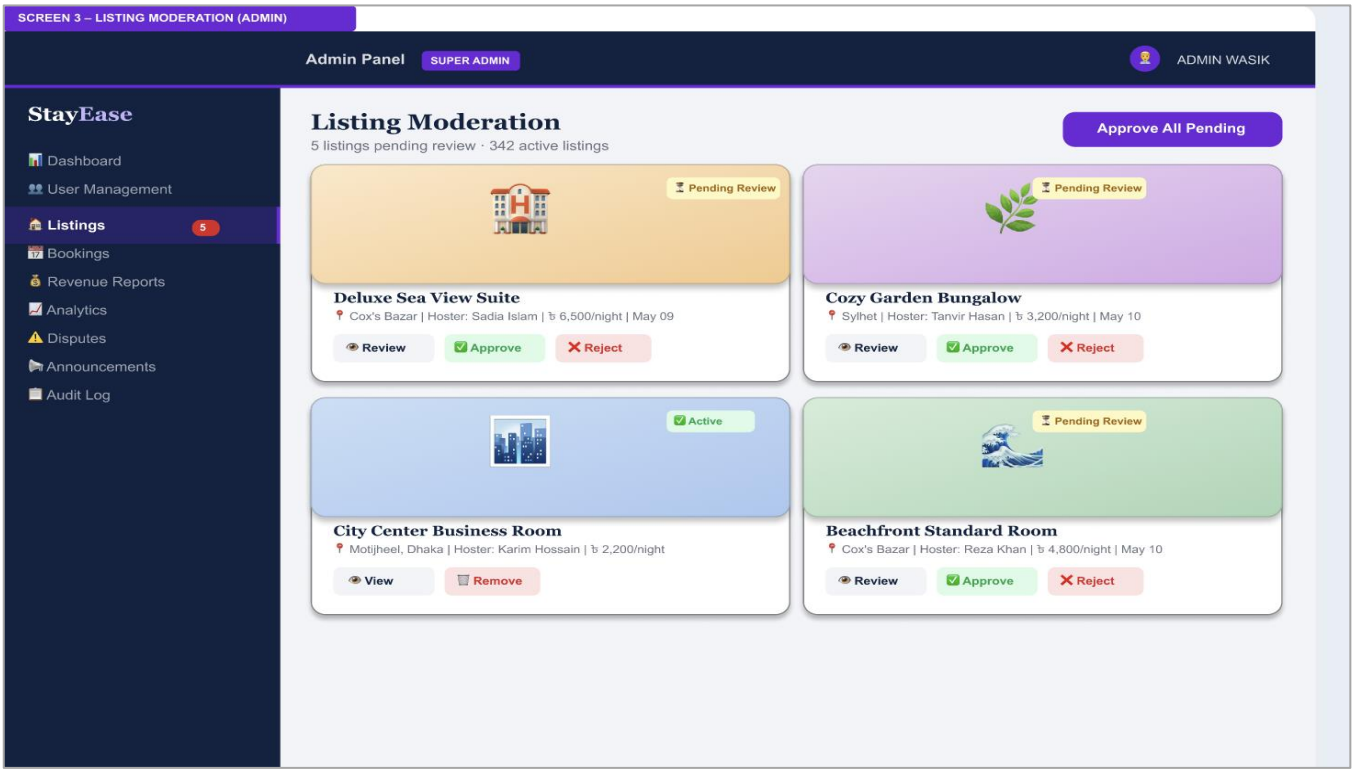


Fig 4.3.3: Listing Moderation (Admin)

Description: This screen presents newly submitted and existing room listings as individual cards for the Admin to moderate. Each card shows the listing's title, location, hosting owner, nightly price, and current status (Pending Review or Active), along with contextual actions — Review, Approve, Reject for pending listings, and View, Remove for already-active ones. A summary line at the top shows how many listings are pending review versus currently active, and an 'Approve All Pending' button offers a bulk-action shortcut. This screen directly implements FR 5.3 and FR 5.4 (Admin approval required before a listing becomes public, and Admin's ability to remove policy-violating listings).

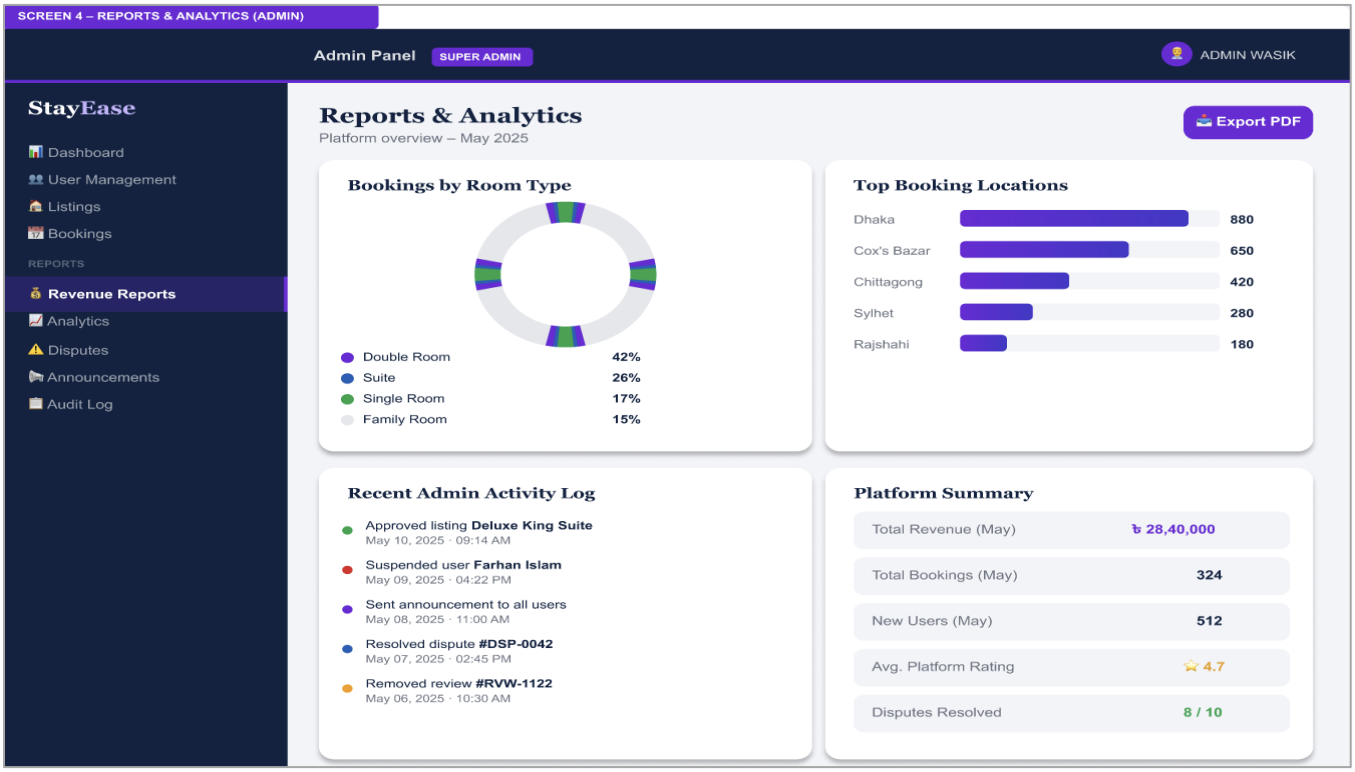


Fig 4.3.4: Report & Analysis (Admin)

Description: This screen gives the Admin an analytical overview of the platform's performance for the selected period. A donut chart breaks down bookings by room type (Double, Suite, Single, Family), while a horizontal bar chart ranks the top booking locations across Bangladesh (Dhaka, Cox's Bazar, Chittagong, Sylhet, Rajshahi). A 'Recent Admin Activity Log' tracks moderation actions such as listing approvals, user suspensions, and dispute resolutions with timestamps, and a 'Platform Summary' panel consolidates total revenue, bookings, new users, average rating, and disputes resolved. An 'Export PDF' button lets the Admin download the full report.



Fig 4.3.5: Homepage

Description: This is the public-facing landing page Guests see when they first visit StayEase. A prominent hero section with the tagline 'Find Your Perfect Stay in Bangladesh' contains the primary search widget, where guests enter a location, check-in/check-out dates, and number of guests. Below the search bar, platform trust indicators (rooms listed, happy guests, cities covered, average rating) build credibility, followed by a 'Featured Rooms' section highlighting handpicked listings with photos, location, rating, and price per night.

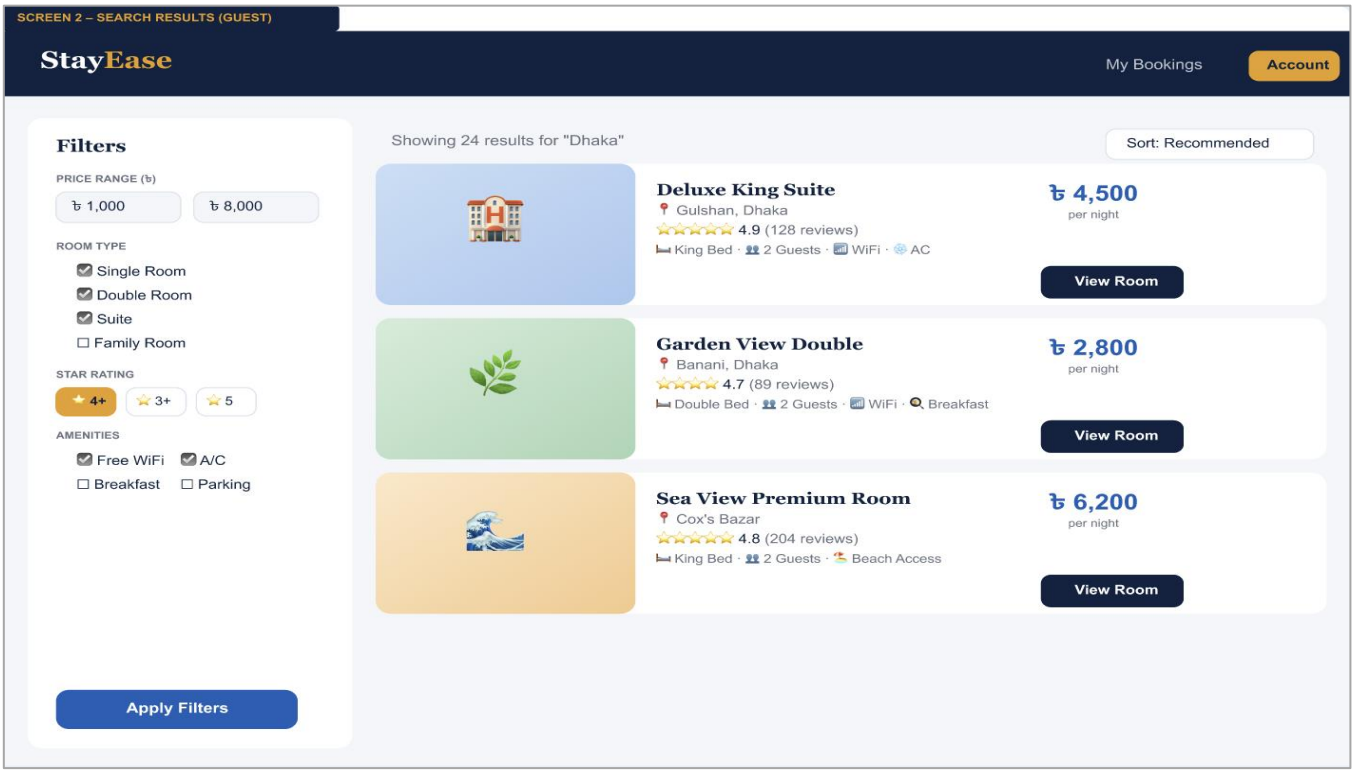


Fig 4.3.6: Search result (Guest)

Description: This screen displays the list of room listings returned for a Guest's search query, here showing 24 results for 'Dhaka'. A left-hand filter panel lets guests narrow results by price range, room type (Single, Double, Suite, Family), minimum star rating, and amenities such as Free WiFi, A/C, Breakfast, and Parking, with an 'Apply Filters' action at the bottom. Each result card in the main list shows a room's title, location, rating with review count, key details (bed type, guest capacity, amenities), and price per night, alongside a 'View Room' button. A sort control in the top-right lets guests reorder results.

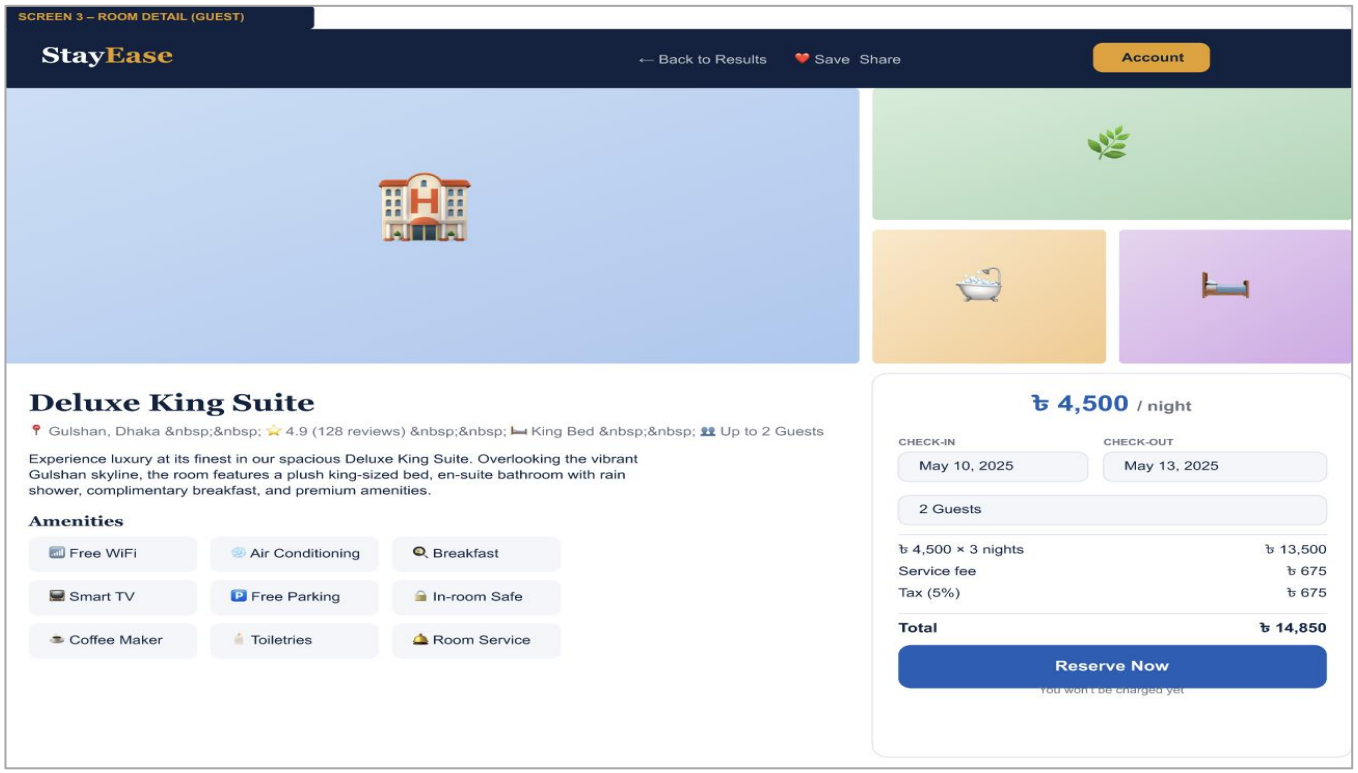


Fig 4.3.7: Room details

Description: This screen shows the full detail page for a single room listing, in this case the 'Deluxe King Suite'. A photo gallery sits at the top, followed by the room title, location, rating, bed configuration, and guest capacity, along with a descriptive summary and a grid of amenity tags (Free WiFi, Air Conditioning, Breakfast, Smart TV, Parking, and more). A sticky booking panel on the right lets the guest confirm check-in/check-out dates and guest count, and displays a transparent cost breakdown — nightly rate × nights, service fee, and tax — culminating in the total price and a 'Reserve Now' button.

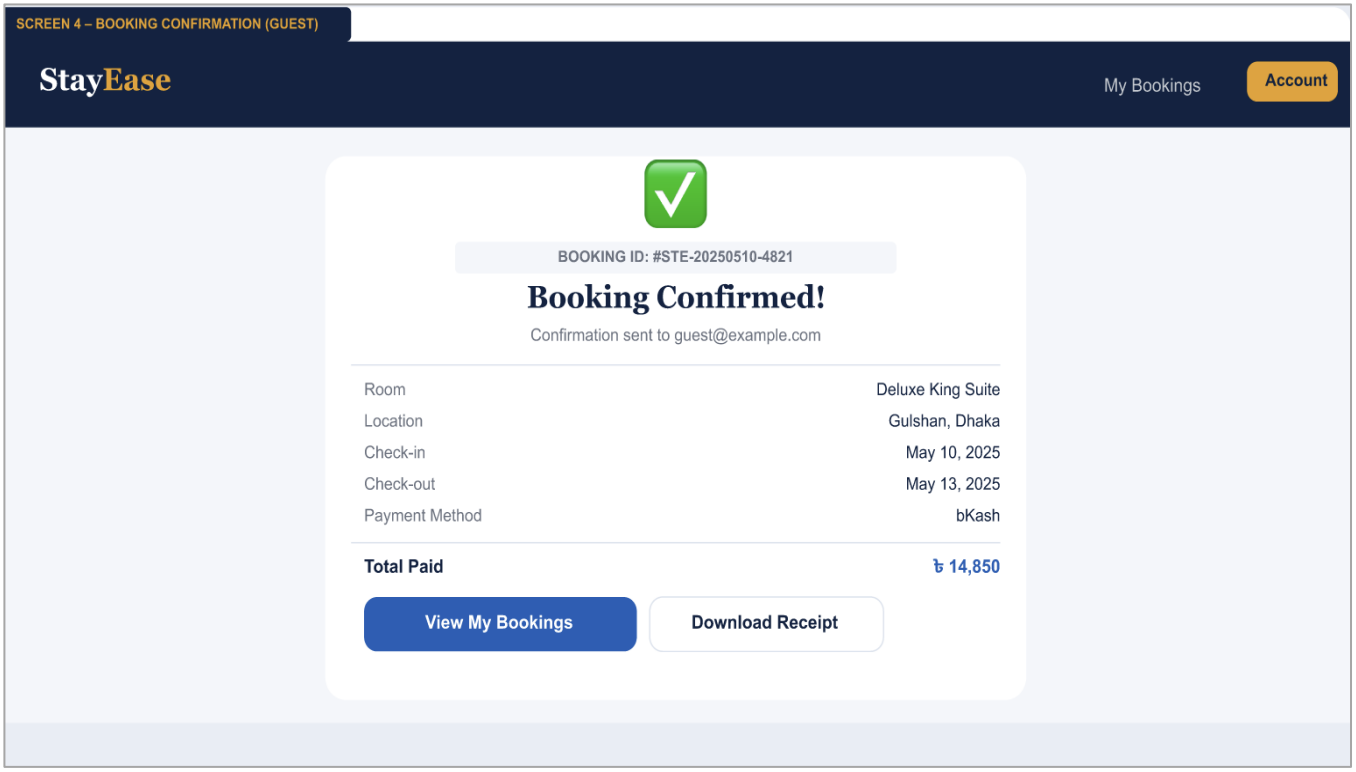


Fig 4.3.8: Booking Confirmation

Description: This screen is shown immediately after a Guest successfully completes payment for a booking. A green checkmark and 'Booking Confirmed!' message reassure the guest that the reservation is complete, alongside the system-generated Booking ID and a note that a confirmation email has been sent to their registered address. A details panel summarizes the room name, location, check-in/check-out dates, payment method used (bKash), and total amount paid. Two buttons below the details panel let the guest navigate to 'View My Bookings' or 'Download Receipt'.

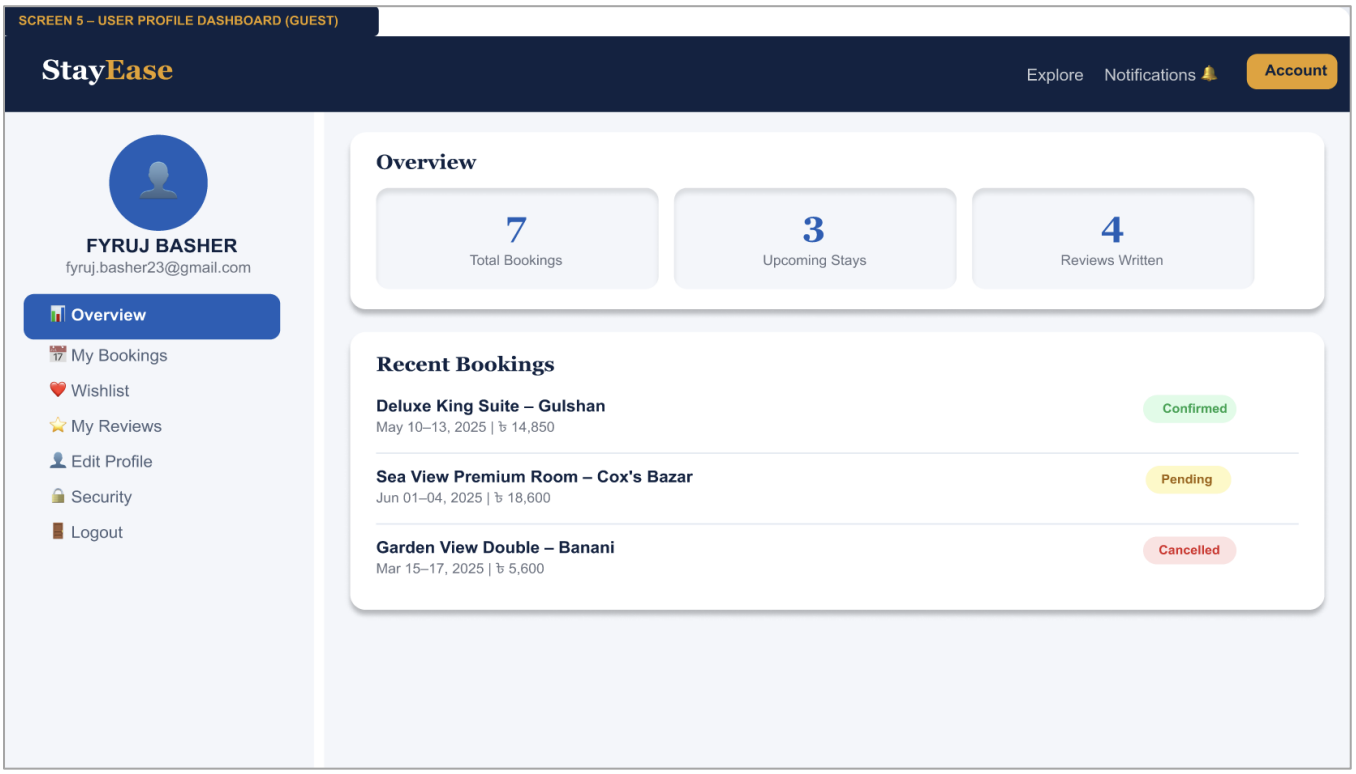


Fig 4.3.9: User Profile (Guest)

Description: This screen serves as the Guest's personal account hub, accessible after login. A profile card on the left displays the guest's name and email alongside navigation links to My Bookings, Wishlist, My Reviews, Edit Profile, Security, and Logout. The main panel opens on an 'Overview' tab showing summary counters for total bookings, upcoming stays, and reviews written, followed by a 'Recent Bookings' list that shows each stay's room name, location, dates, price, and current status (Confirmed, Pending, or Cancelled) using color-coded badges.

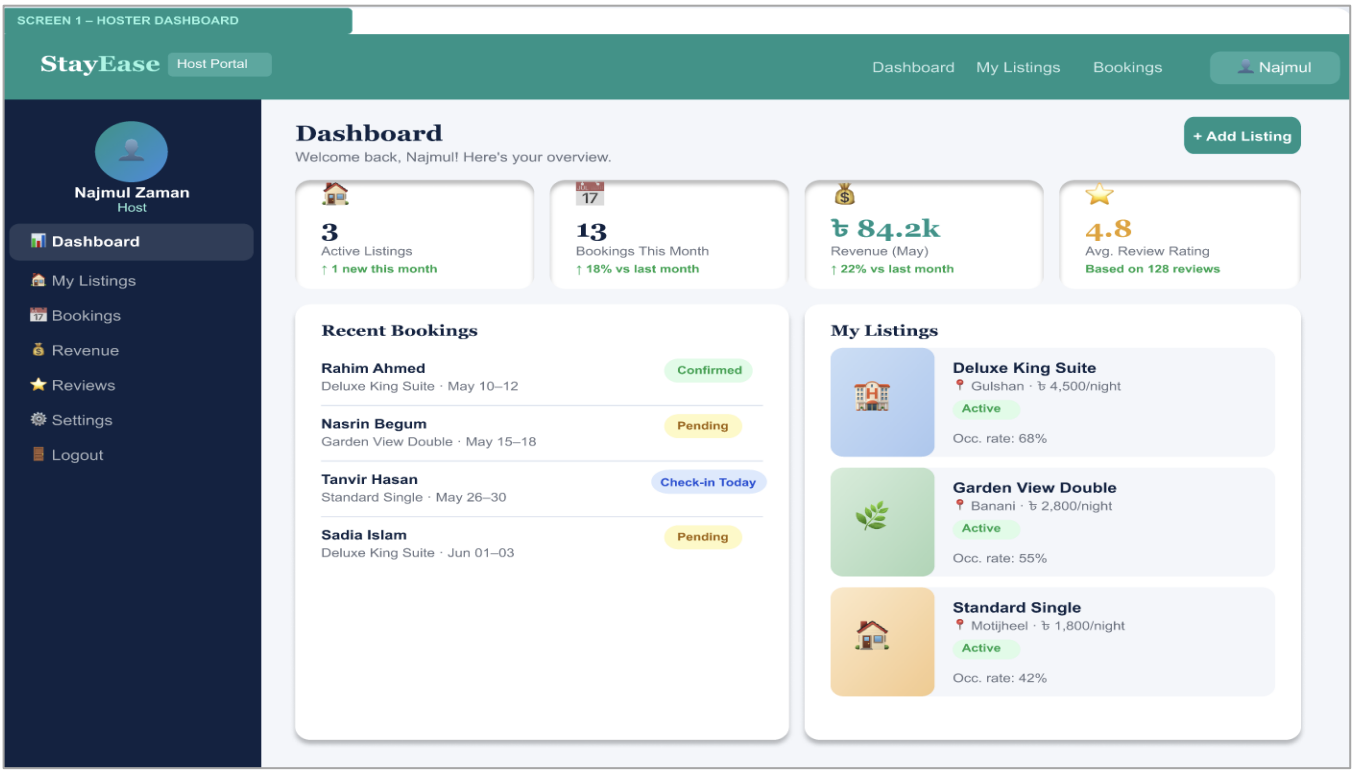


Fig 4.3.10: Host Dashboard

Description: This screen is the main landing page for a logged-in Hotel Room Host, here shown for host 'Najmul Zaman'. Four summary cards display the host's active listings count, bookings this month, monthly revenue, and average review rating, each with a trend indicator. A 'Recent Bookings' panel lists incoming reservations with guest name, room, dates, and status (Confirmed, Pending, Check-in Today), while the 'My Listings' panel shows each of the host's active properties with location, price, status, and occupancy rate. An '+ Add Listing' button in the top-right lets the host quickly create a new listing.

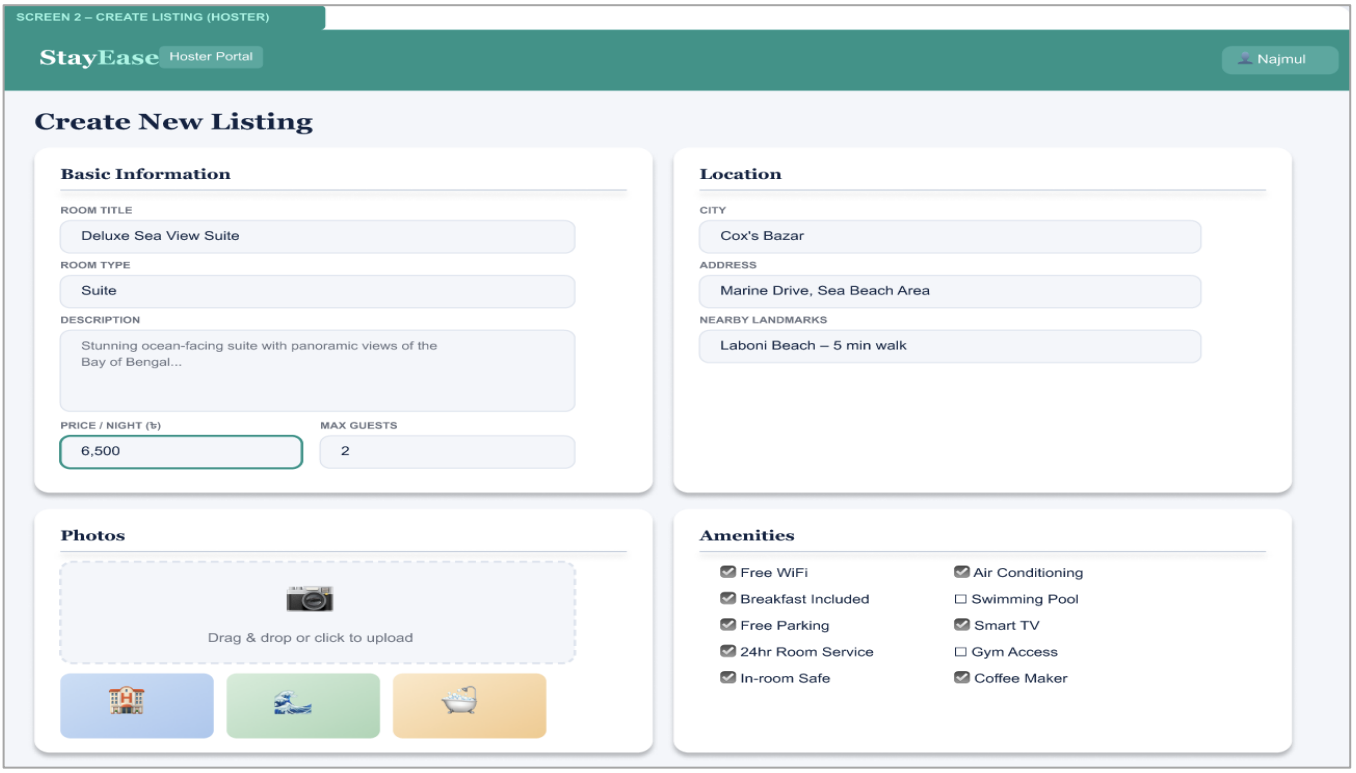


Fig 4.3.11: Create Listing (Host)

Description: This screen provides the form a verified Host uses to publish a new room listing, organized into four clear sections. 'Basic Information' captures the room title, type, description, price per night, and maximum guest capacity, while 'Location' captures the city, address, and nearby landmarks. A 'Photos' section provides drag-and-drop image upload with thumbnail previews, and an 'Amenities' section offers a checklist of features such as Free WiFi, Air Conditioning, Breakfast, Parking, and Smart TV.

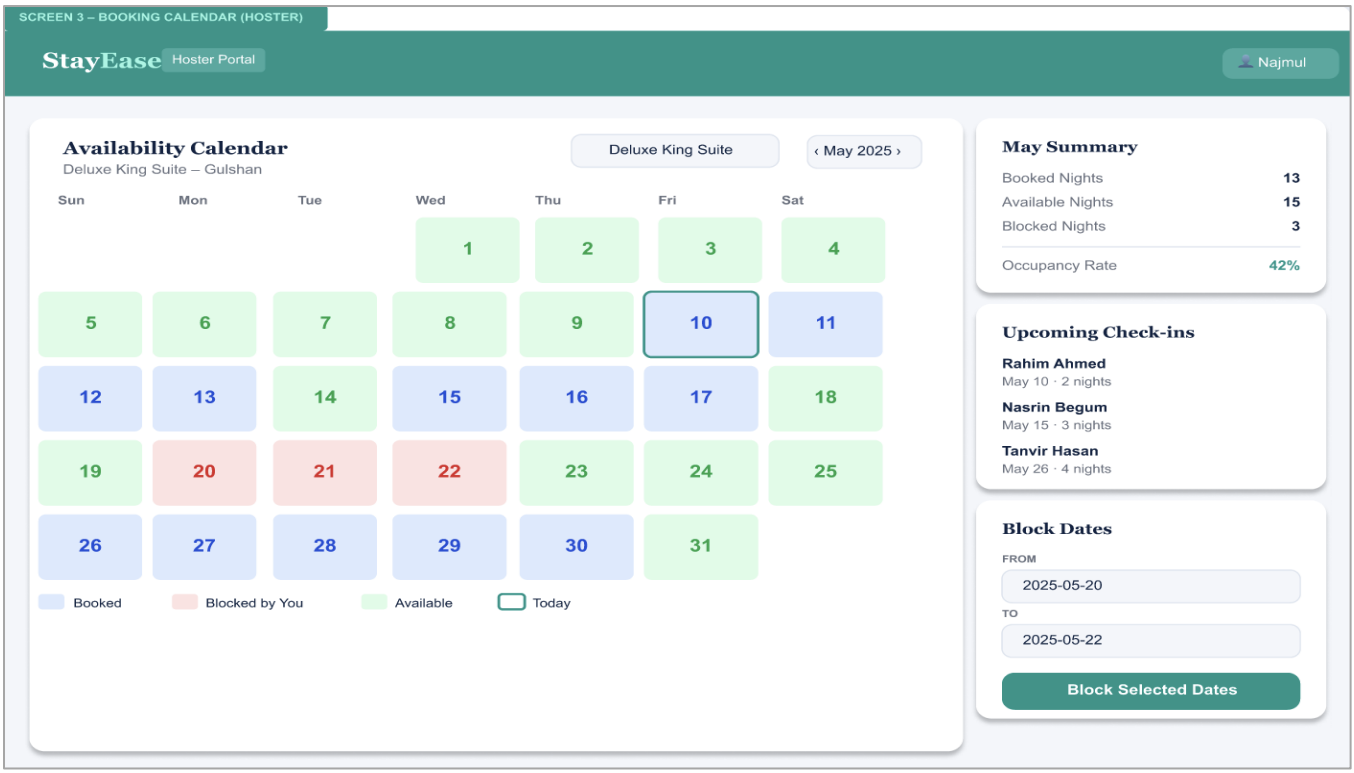


Fig 4.3.12: Booking Calendar (Host)

Description: This screen gives the Host a monthly calendar view of a specific listing's availability, here showing 'Deluxe King Suite – Gulshan' for May 2025. Each date cell is color-coded as Booked, Blocked by You, or Available, with the current day outlined for quick reference. A 'May Summary' panel tallies booked, available, and blocked nights along with the resulting occupancy rate, and an 'Upcoming Check-ins' panel lists guests arriving soon with their stay length. A 'Block Dates' form lets the host select a date range and block it from bookings (e.g., for maintenance).

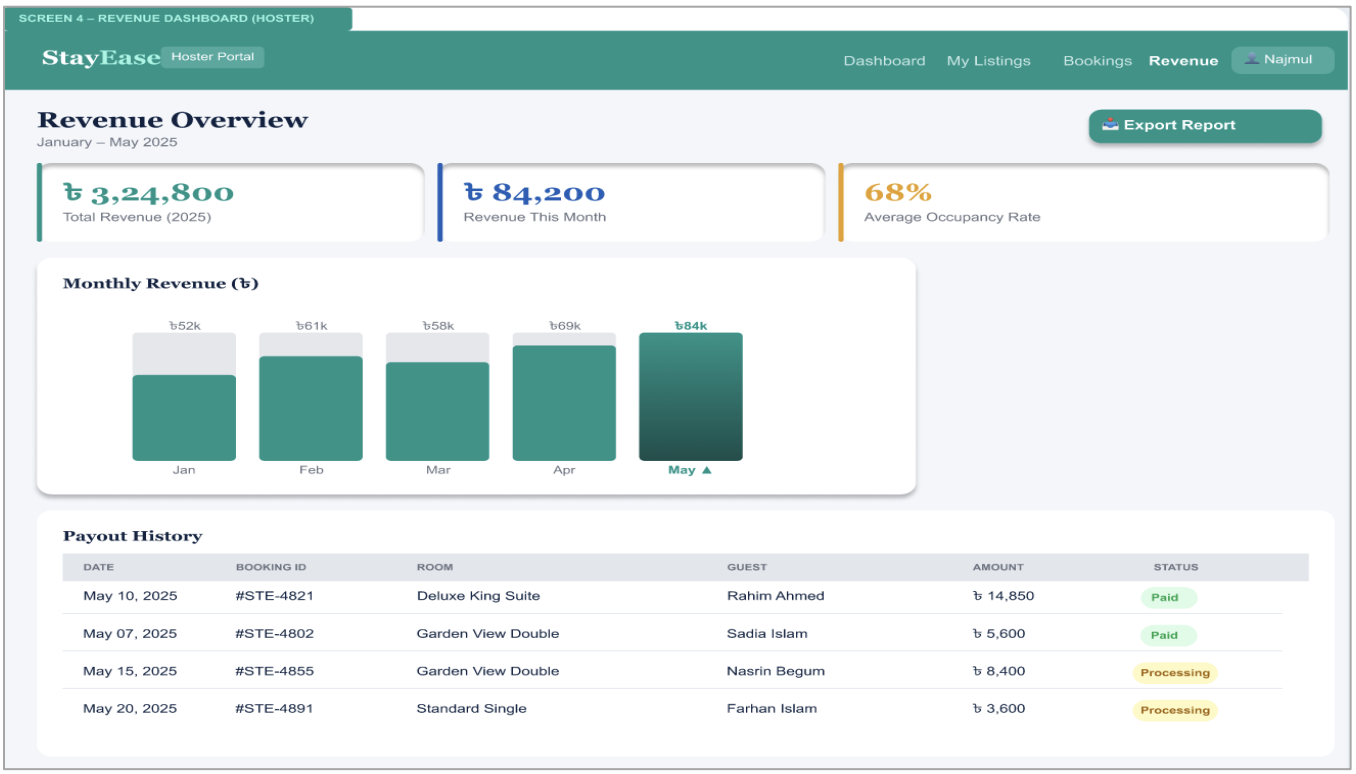


Fig 4.3.13: Revenue Dashboard (Host)

Description: This screen gives the Host a financial overview of their earnings on StayEase, covering January through May 2025. Three summary cards show total revenue for the year, revenue for the current month, and average occupancy rate across the host's listings. A bar chart visualizes month-over-month revenue growth, making seasonal or growth trends easy to spot at a glance. Below the chart, a 'Payout History' table lists individual transactions with date, Booking ID, room, guest name, amount, and payment status (Paid or Processing), and an 'Export Report' button allows the host to download this data.

4.4 Project Requirements

4.4.1 Effort Estimation

Project category is **Semi-Detached**. Because medium size, moderately complex and team have mixed experience.

Software Project Type	Coefficient <Effort Factor>	P	T
Organic	2.4	1.05	0.38
Semi-detached	3.0	1.12	0.35
Embedded	3.6	1.20	0.32

Software Project Type: Semi detached

Estimated SLOC: 9000

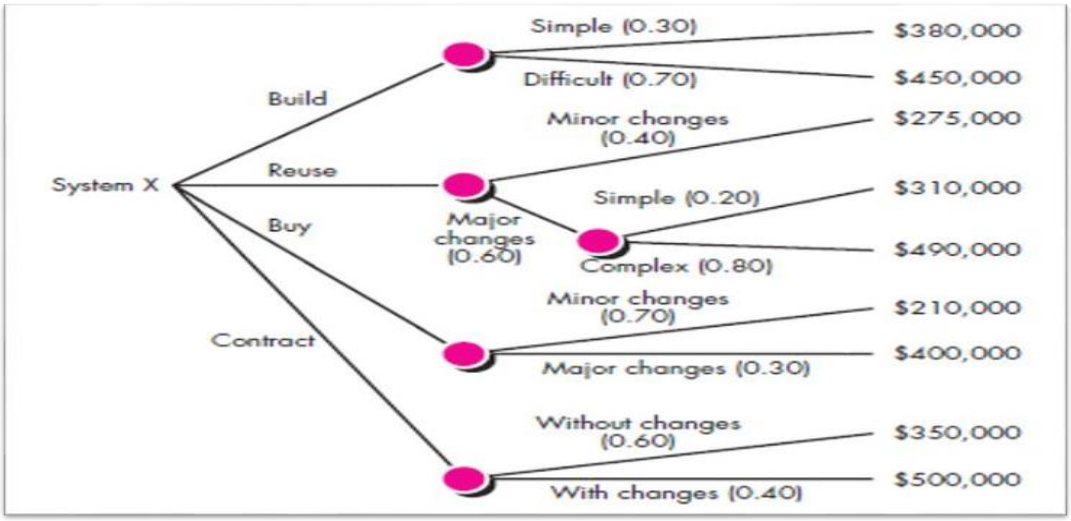
Effort = **PM** = Coefficient<Effort Factor> * (SLOC/1000) ^P = 3.0x(9000/1000) ^1.12 = 35.15 PERSON MONTH

Development Time = **DM** = 2.50 * (PM)^T = 2.50 x (35.15) ^0.35 = **8.69 MONTHS**

Required Number of People = **ST** = PM/DM = 35.15/8.69 = 4.04 ≈ **4 DEVELOPER**

4.4.2 Budget Estimation

Computing expected cost from Decision Tree



the expected cost of **Build** is:

(0.30×380,000) + (0.70×450,000)
= 114,000+315,000 = ***BDT 429,000***

the expected cost of **Reuse** is:

expected cost if there are major changes:

(0.20×310,000) + (0.80×490,000)
= 62,000+392,000 = BDT 454,000

the overall Reuse cost:

(0.40×275,000) + (0.60×454,000)
= 110,000+272,400 = ***BDT 382,400***

the expected cost of **Buy** is:

(0.70×210,000) + (0.30×400,000)
= 147,000+120,000 = ***BDT 267,000***

the expected cost of **Contract** is:

(0.60×350,000) + (0.40×500,000)
= 210,000+200,000 = ***BDT 410,000***

Choice	**Expected cost**
Build	BDT 429,000
Reuse	BDT 382,400
Buy	BDT 267,000
Contract	BDT 410,000

Therefore, based on the decision tree, **Buy** is the best option because it has the lowest expected cost: BDT 267,000.

4.4.3 Resources

1. Human Resources

The following team members and roles are required for the project:

Role	Responsibility
Project Manager	Manage project planning and progress
UI/UX Designer	Design user-friendly interface
Frontend Developer	Develop user interface and mobile/web screens
Backend Developer	Develop APIs, authentication, database handling
Database Administrator	Manage medical records and database security
QA/Test Engineer	Perform software testing and bug tracking
DevOps/Cloud Engineer	Handle deployment and server maintenance

2. Software Resources

Software/Tool	Purpose
Visual Studio Code	Code editor
React / Next js	Frontend Development
Node.js / Django / Laravel	Backend Development
MySQL / PostgreSQL	Database Management
Git & GitHub	Version Control
Twilio	Notifications and OTP
Zoom/WebRTC API	Video consultation

4.4.4 Environment

1. Web browser-based System
2. Cloud deployment environment

The System will be deployed using:

- Cloud hosting platform
- Secure HTTPS/SSL connection
- Real-time database synchronization
- Backup and recovery server

3. Network Environment

- High-speed internet connection
- Secure API communication
- GPS and map integration
- Real-time notification services

5. FEATURES NOT TO BE TESTED

1. Third party Payment gateway internal logic (SSL Commerz, Baksh, Nagad, Rocket, Visa Card, Master Card etc.)
2. Google Maps API internal logic
3. External Email API infrastructure
4. External SMS API
5. MySQL database engine internal logic
6. NID verification service internals
7. User-owned device and browser-specific issues

6. TESTING APPROACH

6.1 Testing Levels

- **Unit Testing:** Tests a small software unit at a time (Registration, Login, Payment etc.). It is performed by developers and follows the white box testing method.
- **System Integration Testing:** This method is putting modules together in an incremental manner after conducting unit testing on module. The specific QA team will be responsible for this testing approach.
- **System Testing:** It is the testing of a complete and fully integrated software product. It should test the functional and non-functional requirements of the software and specific QA team will be responsible for this testing approach.

- **Load Testing:** The QA team will evaluate system performance under expected peak-users load.
- **Security Testing:** The QA team will verify data protection, access control, and vulnerability handle.
- **Acceptance Testing:** This testing approach verifies functionality and usability of the software, and it is performed by Customers/ End users. It is confirmed that the system meets the agreed upon criteria.

6.2 Test Tools

The StayEase project will use several testing tools to ensure software quality, security, and reliability across its Guest, Host, and Admin subsystems. Playwright will be used for functional and automated end-to-end testing of the booking, listing, and dashboard interfaces. Postman will test backend APIs, including room search, booking, and payment-gateway communication with SSL Commerz and bKash. Apache JMeter will perform load and performance testing to validate the system's target of 2,500 concurrent users. OWASP ZAP will be used for security vulnerability testing, particularly around authentication, session management, and data transmission (HTTPS/TLS) and Jira will manage bug tracking and issue reporting throughout the QA cycle.

Testing Type	**Tool**	**Purpose**
Functional Testing	Playwright	Test login, forms, search, emergency requests
API Testing	Postman	Test backend APIs and ambulance services
Performance/Load Testing	Apache JMeter	Test 1000+ concurrent users
Security Testing	OWASP ZAP	Detect security vulnerabilities
Database Testing	MySQL Workbench	Verify database records
Bug Tracking	Jira	Track bugs and project tasks

6.3 Meetings

Regular meetings will be conducted throughout the StayEase project lifecycle to ensure proper communication, project monitoring, risk management, and software quality assurance. Different types of meetings will be arranged based on project needs

Meeting Type	Participants	Frequency	Purpose
Project Initial Meeting	Project Manager, Developers, QA Team, Stakeholders	Once at project start	Discuss project goals, scope, requirements, timeline, and responsibilities
Requirement Review Meeting	Project Manager, Developers, QA Team, Client/Stakeholders	Weekly (during requirement phase)	Review and confirm functional and non-functional requirements
Development Progress Meeting	Project Manager, Frontend & Backend Developers	Weekly	Monitor development progress, task completion, and technical issues
QA/Test Planning Meeting	QA Team, Developers, Project Manager	Before testing phase	Finalize test strategy, test cases, tools, and testing schedule
Bug Review Meeting	QA Team and Developers	Twice per week	Discuss detected bugs, severity, priority, and fix status
Sprint/Iteration Meeting	Entire Development Team	Every 2 weeks	Review completed work and plan next iteration tasks
Security & Risk Review Meeting	QA Team, Security Tester, DevOps Engineer	Monthly	Review security vulnerabilities, risks, and mitigation plans
Client Demonstration Meeting	Stakeholders, Project Manager, Development Team	At milestone completion	Demonstrate project features and gather feedback
Final Acceptance Meeting	Client/End Users, QA Team, Project Manager	Before deployment	Confirm system meets requirements and approve final release

7. TEST CASES/TEST ITEMS

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Wasik Billah
Test Case ID: FR_01	**Test Designed date:** 21-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by: _____**
Module Name: User Authentication & Login	**Test Execution date: _____**
Test Title: Verify Guest login with valid registered email and password	
Description: Verify that a registered and verified Guest can successfully log in using a valid email address and password, and is redirected to the Guest dashboard.	
Precondition (If any): User has already registered, verified their email, and account status is Active	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the StayEase login page. 2. Enter a valid, verified email address. 3. Enter the correct account password. 4. Click the 'Login' button.	Email: rahim.ahmed@gmail.com Password: Rahim@123	Credentials are validated against the database; user is redirected to the Guest home dashboard.	As expected,	
Post Condition: A new session is created for the user and the session timer starts (auto-expires after 30 minutes of inactivity).				

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Wasik Billah
Test Case ID: FR_02	**Test Designed date:** 21-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: User Authentication & Login	**Test Execution date:** _____
Test Title: Verify system behavior on incorrect password entry	
Description: Verify that the system displays an error message	
Precondition (If any): User has a valid registered and verified account.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the login page. 2. Enter a valid registered email address. 3. Enter an incorrect password. 4. Click 'Login'.	Email: karim@email.com Password: WrongPass1 (incorrect)	System displays 'Invalid username or password' error message	As expected,	

Post Condition: No session is created; failed login attempt

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Wasik Billah
Test Case ID: FR_03	**Test Designed date:** 21-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: User Authentication & Login	**Test Execution date:** _____
Test Title: Verify account lockout after 3 consecutive failed login attempts	
Description: Verify that the system temporarily blocks account login for one hour after 3 consecutive failed login attempts	
Precondition (If any): User account exists; 2 consecutive failed login attempts have already been made.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the login page. 2. Enter the valid registered email. 3. Enter an incorrect password (3rd consecutive failed attempt). 4. Click 'Login'.	Email: nasrin@email.com Password: incorrectPass (3rd attempt)	System blocks further login attempts for this account for 1 hour and displays a message: 'Account temporarily locked. Please try again after 1 hour.'	As expected,	

Post Condition: Account status is temporarily set to locked in the database for 1 hour; even correct credentials are rejected during this period.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Wasik Billah
Test Case ID: FR_04	**Test Designed date:** 21-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: User Authentication & Login	**Test Execution date:** _____
Test Title: Verify password length validation during registration	
Description: Verify that the system rejects registration when the password entered is shorter than the required 8-character minimum	
Precondition (If any): User is on the Sign-Up page and has not yet registered with the given email.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the registration page. 2. Enter a unique, valid email address. 3. Enter a password with fewer than 8 characters. 4. Click 'Sign Up' / 'Register'.	Email: newguest01@gmail.com Password: abc12 (5 characters)	System rejects the form submission and displays a validation error: 'Password must be at least 8 characters long.' No account is created.	As expected,	

Post Condition: No new user record is created in the database.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Wasik Billah
Test Case ID: FR_05	**Test Designed date:** 21-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: User Authentication & Login	**Test Execution date:** _____
Test Title: Verify email verification requirement before account activation	
Description: Verify that a newly registered account remains inactive and cannot log in until the user verifies their email via the verification link	
Precondition (If any): User has just completed registration with a valid email and password and has not clicked the verification link yet.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Register a new account with a valid, unique email and a password of at least 8 characters. 2. Do NOT click the verification link sent to the email. 3. Attempt to log in with the same email and password immediately.	Email: sadia.new@gmail.com Password: Sadia@2026	Login attempt is rejected with message: 'Please verify your email before logging in.' Account status remains inactive until verification.	As expected,	
Post Condition: Account remains inactive in the database until the verification link is clicked.				

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Wasik Billah
Test Case ID: FR_06	**Test Designed date:** 21-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: User Authentication & Login	**Test Execution date:** _____
Test Title: Verify 'Forgot Password' generates a time-limited reset link	
Description: Verify that requesting a password reset sends a time-limited reset link to the user's registered email	
Precondition (If any): User has a valid, registered, and verified account.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the login page and click 'Forgot Password'. 2. Enter the registered email address. 3. Click 'Send Reset Link'. 4. Open the received email and click the reset link within the validity window.	Email: farhan.islam@email.com	A password reset email is sent containing a time-limited link; clicking the link within the valid period allows the user to set a new password.	As expected,	

Post Condition: The reset link becomes invalid/expired after use or after the time limit elapses, and the account password is updated.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Wasik Billah
Test Case ID: FR_07	**Test Designed date:** 21-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: User Authentication & Login	**Test Execution date:** _____
Test Title: Verify session auto-expiry after 30 minutes of inactivity	
Description: Verify that a logged-in user's session is automatically invalidated after 30 minutes of inactivity, forcing re-authentication	
Precondition (If any): User is logged in and has an active session.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in successfully as a Guest. 2. Remain idle without any interaction with the system for 30 minutes. 3. After the idle period, attempt to navigate to a protected page (e.g., 'My Bookings').	Idle duration: 30 minutes (no clicks/requests)	The user's session has expired; the system redirects to the login page with a message: 'Session expired, please log in again.'	As expected,	

Post Condition: The session token is invalidated server-side and no longer grants access to protected resources.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Wasik Billah
Test Case ID: FR_08	**Test Designed date:** 21-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: User Authentication & Login	**Test Execution date:** _____
Test Title: Verify Role-Based Access Control (RBAC) blocks unauthorized access	
Description: Verify that a logged-in Host cannot access Admin-only pages/features, enforcing role-based access control	
Precondition (If any): A Host account is logged in successfully.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in with valid Host credentials. 2. Manually enter/navigate to the Admin Dashboard URL (e.g., /admin/dashboard). 3. Observe the system response.	Role: Host Target URL: /admin/dashboard	System denies access and either redirects to the Host dashboard or displays a '403 Forbidden / Access Denied' message. Admin-only data is not exposed.	As expected,	
Post Condition: No Admin-level data or actions are accessible to the Host account.				

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Wasik Billah
Test Case ID: FR_09	**Test Designed date:** 21-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: Room Search and Discovery	**Test Execution date:** _____
Test Title: Verify room search using location, check-in/out dates and guest count	
Description: Verify that the search function returns relevant room listings when a valid location, check-in date, check-out date, and guest count are provided	
Precondition (If any): The platform has active, approved room listings for the given location.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the StayEase homepage. 2. Enter location 'Dhaka'. 3. Select Check-in: 10 May 2026 and Check-out: 13 May 2026. 4. Enter number of guests: 2. 5. Click 'Search'.	Location: Dhaka Check-in: 2026-05-10 Check-out: 2026-05-13 Guests: 2	System displays a list of room listings in Dhaka available for the selected date range that can accommodate 2 guests, each with photo, title, location, price/night, and rating.	As expected,	

Post Condition: Search query and result set are logged for the session; no data is modified.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Wasik Billah
Test Case ID: FR_10	**Test Designed date:** 21-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: Room Search and Discovery	**Test Execution date:** _____
Test Title: Verify only rooms with confirmed availability for selected dates are shown	
Description: Verify that rooms already booked for the selected date range are excluded from search results	
Precondition (If any): At least one room in the search location is already fully booked for the selected date range.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Search rooms in 'Cox's Bazar' for Check-in: 01 Jun 2026, Check-out: 04 Jun 2026. 2. Review the list of returned room listings. 3. Cross-check a known already-booked room ('Sea View Premium Room') against the result list.	Location: Cox's Bazar Check-in: 2026-06-01 Check-out: 2026-06-04	The already-booked room ('Sea View Premium Room') for those dates does NOT appear in the search results; only rooms with confirmed availability are listed.	As expected,	

Post Condition: No booking or availability data is altered.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Jannatul Nayeem
Test Case ID: FR_11	**Test Designed date:** 23-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: Room Search and Discovery	**Test Execution date:** _____
Test Title: Verify filtering search results by price range	
Description: Verify that the guest can filter search results using a minimum and maximum price range	
Precondition (If any): Search results for a location are already displayed with rooms at varying prices.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Perform a room search for 'Dhaka'. 2. In the Filters panel, set Price Range from ৳1,000 to ৳3,000. 3. Click 'Apply Filters'.	Price Range: ৳1,000 - ৳3,000	Only room listings priced between ৳1,000 and ৳3,000 per night are displayed; listings outside this range (e.g., ৳4,500/night) are hidden.	As expected,	

Post Condition: The filter selection persists while the guest continues browsing within the same search session.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Jannatul Nayeem
Test Case ID: FR_12	**Test Designed date:** 23-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: Room Search and Discovery	**Test Execution date:** _____
Test Title: Verify filtering search results by room type and amenities	
Description: Verify that search results can be filtered by room type (Single, Double, Suite, Family) and amenities (e.g., Free WiFi, A/C)	
Precondition (If any): Search results are displayed for a chosen location and date range.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Perform a room search for 'Dhaka'. 2. In Filters, select Room Type = 'Suite' and Amenities = 'Free WiFi' and 'A/C'. 3. Click 'Apply Filters'.	Room Type: Suite Amenities: Free WiFi, A/C	Only Suite-type listings that offer both Free WiFi and A/C are displayed in the result list.	As expected,	

Post Condition: Filter state remains applied until cleared or a new search is performed.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Jannatul Nayeem
Test Case ID: FR_13	**Test Designed date:** 23-Aug-2026
Test Priority (Low, Medium, High): Low	**Test Executed by:** _____
Module Name: Room Search and Discovery	**Test Execution date:** _____
Test Title: Verify sorting search results by price (low to high)	
Description: Verify that guests can sort search results by price in ascending order	
Precondition (If any): Multiple room listings with different prices are returned for a search.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Perform a room search returning at least 3 listings with different prices. 2. Select the 'Sort by: Price (Low to High)' option. 3. Observe the order of displayed listings.	Sort option: Price - Low to High	Listings are re-ordered so that the lowest-priced room per night appears first and the highest-priced room appears last.	As expected,	

Post Condition: No underlying listing data changes; only the display order is affected.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Jannatul Nayeem
Test Case ID: FR_14	**Test Designed date:** 23-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: Room Booking and Payment	**Test Execution date:** _____
Test Title: Verify Guest can initiate a booking for an available room	
Description: Verify that a logged-in Guest can initiate a booking by selecting valid check-in/check-out dates and confirming guest count	
Precondition (If any): Guest is logged in; selected room ('Deluxe King Suite') is available for the chosen dates.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as a Guest and open the 'Deluxe King Suite' room detail page. 2. Select Check-in: 10 May 2026, Check-out: 13 May 2026. 3. Set number of guests to 2. 4. Click 'Reserve Now'.	Room: Deluxe King Suite Check-in: 2026-05-10 Check-out: 2026-05-13 Guests: 2	System proceeds to the booking summary/payment step showing the selected dates, guest count, and room details without error.	As expected,	

Post Condition: A pending booking record is created in the system with status 'Pending' until payment is completed.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Jannatul Nayeem
Test Case ID: FR_15	**Test Designed date:** 23-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: Room Booking and Payment	**Test Execution date:** _____
Test Title: Verify total booking cost calculation before payment	
Description: Verify that the system correctly calculates and displays the total cost (nightly rate + service fee + tax) before payment confirmation	
Precondition (If any): Guest has selected a room and valid check-in/check-out dates on the booking page.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open 'Deluxe King Suite' priced at ৳4,500/night. 2. Select Check-in: 10 May 2026, Check-out: 13 May 2026 (3 nights). 3. Proceed to the booking summary screen. 4. Review the displayed cost breakdown.	Nightly rate: ৳4,500 x 3 nights Service fee: ৳675 Tax (5%): ৳675	System displays: Room subtotal ৳13,500 + Service fee ৳675 + Tax ৳675 = Total ৳14,850, matching the correct arithmetic sum before payment confirmation.	As expected,	

Post Condition: No charge occurs yet; the calculated total is carried forward to the payment step.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Jannatul Nayeem
Test Case ID: FR_16	**Test Designed date:** 23-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: Room Booking and Payment	**Test Execution date:** _____
Test Title: Verify successful payment via bKash confirms booking and updates availability	
Description: Verify that upon successful bKash payment, the system marks the room unavailable for the booked dates, generates a unique Booking ID, and sends a confirmation email	
Precondition (If any): Guest has a pending booking with total amount calculated and is on the payment screen.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. On the payment screen, select 'bKash' as the payment method. 2. Enter valid bKash account/PIN details in the test payment gateway. 3. Confirm the payment. 4. Wait for the confirmation screen to load.	Payment method: bKash Amount: ৳14,850 Account: 01700000000	Payment is processed successfully; system displays 'Booking Confirmed' with a unique Booking ID (e.g., #STE-20260510-4821); the room's calendar is updated to unavailable for 10-13 May 2026; a confirmation email is sent to the guest's registered email.	As expected,	

Post Condition: Booking status changes from 'Pending' to 'Confirmed' in the database; availability calendar and host notifications are updated.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Jannatul Nayeem
Test Case ID: FR_17	**Test Designed date:** 23-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: Room Booking and Payment	**Test Execution date:** _____
Test Title: Verify failed payment does not create a confirmed booking	
Description: Verify that if payment fails or is declined, the booking is NOT confirmed, the room remains available, and no partial charge occurs, satisfying the atomic transaction requirement	
Precondition (If any): Guest has a pending booking and reaches the payment screen; a payment failure is simulated (e.g., insufficient balance/declined transaction).	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. On the payment screen, select 'bKash' as payment method. 2. Enter test credentials configured to simulate a declined/failed transaction. 3. Submit the payment. 4. Observe the system response and check the room's availability afterward.	Payment method: bKash Amount: ৳14,850	System displays 'Payment Failed' with an option to 'Try Payment Again'; the booking status remains 'Pending' (not confirmed); the room's availability calendar is NOT changed to unavailable; no confirmation email is sent.	As expected,	

Post Condition: Room remains available for other guests to book the same dates; no financial transaction record with 'Paid' status exists for this attempt.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Jannatul Nayeem
Test Case ID: FR_18	**Test Designed date:** 23-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: Room Booking and Payment	**Test Execution date:** _____
Test Title: Verify system prevents double-booking of the same room and dates	
Description: Verify that once a room is booked/paid for specific dates, a second guest cannot book the same room for overlapping dates (Ref. SRS FR 3.4, Business Objective: eliminate double-booking).	
Precondition (If any): 'Garden View Double' room is already confirmed and booked for 15-18 May 2026 by Guest A.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as Guest B (a different account). 2. Search for and open 'Garden View Double' room detail page. 3. Attempt to select Check-in: 16 May 2026, Check-out: 17 May 2026 (overlapping with Guest A's booking). 4. Attempt to proceed with booking.	Room: Garden View Double Check-in: 2026-05-16 Check-out: 2026-05-17 (overlaps existing booking)	System either excludes these dates from the available calendar or displays an error: 'Selected dates are not available for this room.' Booking is blocked for Guest B.	As expected,	

Post Condition: No duplicate/conflicting booking record is created for the same room and overlapping dates.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Jannatul Nayeem
Test Case ID: FR_19	**Test Designed date:** 23-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: Room Booking and Payment	**Test Execution date:** _____
Test Title: Verify booking cancellation follows the host's defined cancellation policy	
Description: Verify that a Guest can cancel a confirmed booking and that the system applies the applicable cancellation policy (Flexible, Moderate, or Strict) (Ref. SRS FR 3.6).	
Precondition (If any): Guest has a confirmed booking for a room with a 'Moderate' cancellation policy, and cancellation is requested outside the free-cancellation window.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as the Guest who made the booking. 2. Navigate to 'My Bookings' and select the confirmed booking. 3. Click 'Cancel Booking'. 4. Confirm the cancellation request.	Booking: Garden View Double (Moderate cancellation policy) Cancellation date: 2 days before check-in	System calculates the refund/penalty per the Moderate policy rules, updates the booking status to 'Cancelled', and displays the applicable refund amount to the guest.	As expected,	

Post Condition: Booking status is updated to 'Cancelled' in the database; the room's availability calendar is updated to reopen the previously blocked dates.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Md. Jannatul Nayeem
Test Case ID: FR_20	**Test Designed date:** 23-Aug-2026
Test Priority (Low, Medium, High): Low	**Test Executed by: _____**
Module Name: Room Booking and Payment	**Test Execution date: _____**
Test Title: Verify Guest can view complete booking history in profile dashboard	
Description: Verify that a Guest can view a complete, accurate history of all their bookings (confirmed, pending, and cancelled) in their profile dashboard	
Precondition (If any): Guest account has at least 3 bookings with different statuses (Confirmed, Pending, Cancelled).	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as the Guest (e.g., Fyruj Basher). 2. Navigate to the Profile Dashboard. 3. Open the 'My Bookings' / 'Recent Bookings' section. 4. Review the listed bookings and their statuses.	Account: fyruj.basher23@gmail.com	All bookings made by the guest are listed with correct room name, dates, amount, and current status (Confirmed / Pending / Cancelled), matching the actual booking records.	As expected,	

Post Condition: No data is modified; this is a read-only view.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Kingkor Karmoker
Test Case ID: FR_21	**Test Designed date:** 24-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: Room Listing Management (Host)	**Test Execution date:** _____
Test Title: Verify verified Host can create a new room listing with all required fields	
Description: Verify that a verified Host can successfully create a new room listing by providing title, description, location, room type, amenities, pricing, and photos	
Precondition (If any): Host account is registered, email-verified, and approved by Admin.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as a verified Host. 2. Navigate to 'My Listings' > 'Add Listing'. 3. Fill in Room Title, Room Type, Description, Price/Night, Max Guests, Location/Address, and select Amenities. 4. Upload at least one valid photo and click 'Save' / 'Publish Listing'.	Title: Deluxe Sea View Suite Room Type: Suite Price: ৳6,500/night Max Guests: 2 Location: Cox's Bazar, Marine Drive	The listing is saved successfully with status 'Pending Review' and appears in the Host's 'My Listings' page, awaiting Admin approval before becoming publicly visible.	As expected,	

Post Condition: A new listing record is created in the database with status = PENDING, linked to the Host's account.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Kingkor Karmoker
Test Case ID: FR_22	**Test Designed date:** 24-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: Room Listing Management (Host)	**Test Execution date:** _____
Test Title: Verify system enforces image upload limits (max 10 images, 5MB each, JPG/PNG only)	
Description: Verify that the system rejects image uploads that exceed 10 images per listing, exceed 5MB per file, or are not in JPG/PNG format	
Precondition (If any): Host is on the 'Create/Edit Listing' page for a room listing.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Attempt to upload 11 valid JPG images to a single listing. 2. Attempt to upload a single JPG image of 7MB in size. 3. Attempt to upload a file in .gif format. 4. Observe system validation messages for each attempt.	Attempt 1: 11 x JPG images (each <5MB) Attempt 2: 1 x JPG image, 7MB Attempt 3: 1 x .gif image	System rejects the 11th image with 'Maximum 10 images allowed per listing'; rejects the 7MB file with 'File size exceeds 5MB limit'; rejects the .gif file with 'Only JPG/PNG formats are supported'.	As expected,	

Post Condition: Only valid images (≤10 count, ≤5MB, JPG/PNG) are actually stored and associated with the listing.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Kingkor Karmoker
Test Case ID: FR_23	**Test Designed date:** 24-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: Room Listing Management (Host)	**Test Execution date:** _____
Test Title: Verify Host can block/open specific dates on the availability calendar	
Description: Verify that a Host can use the availability calendar to block specific dates (e.g., for maintenance) so they cannot be booked by guests	
Precondition (If any): Host is logged in and has an active listing ('Deluxe King Suite') with an availability calendar.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as Host and navigate to 'Availability Calendar' for 'Deluxe King Suite'. 2. Select the date range 20-22 May 2026. 3. Click 'Block Selected Dates'. 4. Log out and, as a Guest, attempt to search/book the same room for 20-22 May 2026.	Room: Deluxe King Suite Block dates: 2026-05-20 to 2026-05-22	The selected dates are marked as 'Blocked by You' on the Host calendar; when a Guest searches for those dates, the room does not appear as available / booking is not possible for that range.	As expected,	

Post Condition: The room's availability calendar reflects the blocked dates until the Host manually reopens them.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Kingkor Karmoker
Test Case ID: FR_24	**Test Designed date:** 24-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: Room Listing Management (Host)	**Test Execution date:** _____
Test Title: Verify Host revenue dashboard displays accurate earnings, booking count, and occupancy rate	
Description: Verify that the Host revenue dashboard correctly reflects total earnings, number of bookings, and monthly occupancy rate based on actual completed bookings	
Precondition (If any): Host has at least 3 completed/confirmed bookings in the current month across their listings.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as Host (e.g., Najmul Zaman). 2. Navigate to the 'Revenue' dashboard. 3. Compare the displayed 'Total Revenue', 'Bookings This Month', and 'Occupancy Rate' against the known confirmed bookings and payout history. 4. Cross-check individual payout entries in the 'Payout History' table.	Host: Najmul Zaman Month: May 2026	Displayed figures (e.g., Total Revenue ৳84,200, Bookings This Month = 13, Occupancy Rate = 68%) match the sum/count of actual confirmed and paid bookings for that host and period.	As expected,	

Post Condition: No data is altered; the dashboard remains a read-only, accurate reporting view.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Kingkor Karmoker
Test Case ID: FR_25	**Test Designed date:** 24-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: Admin Dashboard and Platform Management	**Test Execution date:** _____
Test Title: Verify a newly submitted Host listing requires Admin approval before becoming publicly visible	
Description: Verify that a listing created by a Host is not visible to Guests in search results until it is approved by an Admin	
Precondition (If any): A Host has submitted a new listing that is in 'Pending Review' status.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. As a Guest (or anonymous visitor), search for rooms in the listing's location before Admin approval. 2. Confirm the pending listing does not appear in results. 3. Log in as Admin, navigate to 'Listing Moderation', and click 'Approve' on the pending listing. 4. As a Guest, repeat the same search.	Listing: Beachfront Standard Room (status: Pending Review)	Before approval, the listing does not appear in Guest search results. After the Admin approves it, the listing's status changes to 'Active/Approved' and it becomes visible and bookable in Guest search results.	As expected,	

Post Condition: Listing status is updated from PENDING to APPROVED in the database and becomes publicly searchable.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Kingkor Karmoker
Test Case ID: FR_26	**Test Designed date:** 24-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: Admin Dashboard and Platform Management	**Test Execution date:** _____
Test Title: Verify Admin can suspend a user account for policy violations	
Description: Verify that an Admin can view and suspend any User or Host account, immediately restricting their platform access	
Precondition (If any): Admin is authenticated with admin-level credentials; a target user account ('Farhan Islam') is currently Active.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as Admin and navigate to 'User Management'. 2. Search for the user 'Farhan Islam'. 3. Click 'Suspend' next to the user's record. 4. Confirm the suspension action. 5. Attempt to log in as 'Farhan Islam' using their normal credentials.	Target user: Farhan Islam (farhan@email.com)	User status updates to 'Suspended' in the Admin dashboard; the suspended user's subsequent login attempt is rejected with a message such as 'Your account has been suspended. Contact support.'	As expected,	

Post Condition: User's account status is set to Suspended in the database; the user cannot access any Guest/Host features until reinstated by an Admin.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Kingkor Karmoker
Test Case ID: FR_27	**Test Designed date:** 24-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: Admin Dashboard and Platform Management	**Test Execution date:** _____
Test Title: Verify Admin can generate and export platform reports as PDF/Excel	
Description: Verify that an Admin can generate reports on user activity, revenue, and booking trends and export them in PDF or Excel format	
Precondition (If any): Admin is logged in; sufficient booking/user/revenue data exists for the reporting period (e.g., May 2026).	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as Admin and navigate to 'Reports & Analytics'. 2. Select the reporting period 'May 2026'. 3. Click 'Export PDF' (and separately test 'Export Excel' if available). 4. Open the downloaded file and verify its contents.	Report period: May 2026 Export format: PDF	A PDF (or Excel) file is generated and downloaded successfully, containing accurate platform metrics (total revenue, total bookings, new users, disputes resolved) matching the on-screen dashboard figures.	As expected,	

Post Condition: No underlying data is modified; the exported file is available for the Admin to save/share.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Kingkor Karmoker
Test Case ID: FR_28	**Test Designed date:** 24-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: Review and Rating System	**Test Execution date:** _____
Test Title: Verify Guest can only submit a review after their checkout date has passed	
Description: Verify that the system only allows a Guest to submit a star rating and written review for a room after the checkout date of their stay has passed	
Precondition (If any): Guest has an upcoming (not yet checked-out) confirmed booking, and separately a completed (checked-out) booking, for comparison.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as the Guest and navigate to the booking that has NOT yet reached its checkout date. 2. Attempt to submit a star rating and written review for that booking. 3. Navigate to a different, completed booking whose checkout date has passed. 4. Attempt to submit a review for that completed booking.	Booking A: Checkout date in the future (e.g., 2026-09-15) Booking B: Checkout date already passed (e.g., 2026-05-13)	For Booking A, the 'Submit Review' option is disabled/hidden with a message such as 'You can review this stay after checkout.' For Booking B, the review form is available and the submission succeeds.	As expected,	

Post Condition: A review record is created only for the completed booking (Booking B); no review is recorded against Booking A.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Kingkor Karmoker
Test Case ID: FR_29	**Test Designed date:** 29-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: Notification System	**Test Execution date:** _____
Test Title: Verify booking confirmation email and 24-hour check-in reminder are sent	
Description: Verify that the system sends an automated booking confirmation email immediately after a successful payment, and a separate check-in reminder email 24 hours before the guest's arrival date	
Precondition (If any): Guest has a confirmed, paid booking with a valid registered email address, and the check-in date is exactly 24 hours away.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Complete a room booking and successful bKash/SSL Commerz payment as a Guest. 2. Check the registered email inbox immediately after payment confirmation. 3. Wait until the system clock reaches 24 hours before the booking's check-in date (or trigger the scheduled notification job in the test environment). 4. Check the registered email inbox again for the check-in reminder.	Guest email: guest.test@gmail.com Booking check-in date: 2026-09-10 Current test time: 2026-09-09 (24 hours before check-in)	A booking confirmation email is received immediately after payment, containing Booking ID, room name, and stay dates. A separate check-in reminder email is received exactly 24 hours before the check-in date/time.	As expected,	

Post Condition: Both notification events (booking confirmation, check-in reminder) are logged against the Booking ID in the system's notification history.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Kingkor Karmoker
Test Case ID: FR_30	**Test Designed date:** 29-Aug-2026
Test Priority (Low, Medium, High): Medium	**Test Executed by:** _____
Module Name: Room Listing Management (Host)	**Test Execution date:** _____
Test Title: Verify Host can deactivate a listing, removing it from public search results	
Description: Verify that when a Host deactivates one of their room listings, it is immediately removed from Guest-facing search results while remaining accessible in the Host's own listing management view	
Precondition (If any): Host is logged in and owns at least one active, publicly visible, approved listing ('Standard Single').	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as the Host and navigate to 'My Listings'. 2. Select the 'Standard Single' listing and click 'Deactivate Listing'. 3. Confirm the deactivation action. 4. As a Guest (in a separate session), search for rooms in the same location/type and check whether 'Standard Single' appears.	Listing: Standard Single (Motijheel, Dhaka) Action: Deactivate	The listing's status changes to 'Inactive' in the Host's dashboard; the listing no longer appears in Guest search results or is directly bookable, but remains visible/editable within the Host's own 'My Listings' page for future reactivation.	As expected,	

Post Condition: The listing record's status is updated to INACTIVE in the database; any existing confirmed bookings for dates prior to deactivation remain unaffected.

Project Name: StayEase – Hotel Room Booking Management System	**Test Designed by:** Kingkor Karmoker
Test Case ID: NFR_01	**Test Designed date:** 24-Aug-2026
Test Priority (Low, Medium, High): High	**Test Executed by:** _____
Module Name: Non-Functional Requirement - Performance	**Test Execution date:** _____
Test Title: Verify room search results are returned within 2 seconds under normal load	
Description: Verify that the system returns room search results within 2 seconds under normal operating load, in accordance with the performance requirement	
Precondition (If any): The system is deployed in a test/staging environment simulating normal operating load (up to 2,500 concurrent users) using a load-testing tool (e.g., JMeter).	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Configure the load-testing tool to simulate 2,500 concurrent virtual users browsing the platform. 2. Have a subset of virtual users perform a room search request for 'Dhaka' with a valid date range. 3. Measure the response time from search request submission to results being returned. 4. Record and analyze the response time metrics across multiple iterations.	Concurrent users: 2,500 Search query: Location = Dhaka, valid date range	At least 95% of search requests return results within 2 seconds, and full page loads complete within 3 seconds, with no measurable degradation in user experience, as required by QA2.	As expected,	

Post Condition: Performance metrics (average, 95th percentile response time) are logged for the QA report; no permanent data changes result from the test.

8. ITEM PASS/FAIL CRITERIA

The StayEase system testing process will follow specific pass/fail criteria to determine whether each test item satisfies the expected requirements. A test item will be marked as "Pass" only if the actual result matches the expected result without any critical issues. Otherwise, the test item will be marked as "Fail".

Criteria Type	Pass Criteria	Fail Criteria
Functional Testing	All functions work according to system requirements	Functions produce incorrect or unexpected output
Login & Authentication	Valid users can log in securely and invalid users are rejected	Unauthorized access is possible or valid login fails
Emergency Services	Ambulance and hospital services respond within required time	Delayed response or request failure
Performance Testing	System handles at least 1000 concurrent users without crash	System slowdown, crash, or timeout occurs
Security Testing	No critical vulnerabilities found; encryption works properly	Data leakage, unauthorized access, or security flaws found
Database Testing	Data is stored, updated, and retrieved correctly	Data corruption or incorrect database operations occur
Usability Testing	Users can complete tasks easily without confusion	Navigation problems or unclear interface detected
Compatibility Testing	System works properly on supported browsers/devices	UI breaks or features fail on supported devices
Acceptance Testing	Client confirms requirements are satisfied	Client rejects functionality or requirements are incomplete

Test Completion Criteria:

The testing phase will be considered successfully completed when:

- At least 95% of all test cases pass successfully.
- No critical or high-severity defects remain unresolved.
- All major functional requirements are verified.

- Security and performance requirements meet expected standards.
- Client acceptance testing is approved.

9. TEST DELIVERABLES

The following documents and artefacts will be produced and formally delivered as part of the StayEase software testing process. All deliverables must be reviewed and signed off by the QA Team Lead before release.

Deliverable	Description
Software Test Plan	Complete testing strategy, scope, schedule, and testing process
Test Cases/Test Items	Detailed functional and non-functional test cases
Test Data	Input values and datasets used for testing
Bug/Defect Reports	Reports containing detected issues and bug severity
Test Execution Reports	Results of executed test cases
Performance Test Report	Load and performance testing outcomes
Security Test Report	Vulnerability and security assessment results
Meeting Minutes (MoM)	Records of project and testing meetings
Requirement Traceability Matrix (RTM)	Mapping between requirements and test cases
Final Test Summary Report	Overall testing summary and project quality status
User Acceptance Test (UAT) Report	End-user/client acceptance results

10. STAFFING AND TRAINING NEEDS

The StayEase project requires skilled team members and proper training to ensure effective development and testing activities.

10.1 Staffing Requirements

Role	Required Number	Responsibility
Project Manager	1	Project planning and management
Frontend Developer	3	User interface development
Backend Developer	3	API and server-side development
UI/UX Designer	1	Design application interface
Database Administrator	1	Database management and security
QA/Test Engineer	3	Software testing and bug reporting
DevOps/Cloud Engineer	1	Deployment and server maintenance
Technical Support Staff	1	User and system support

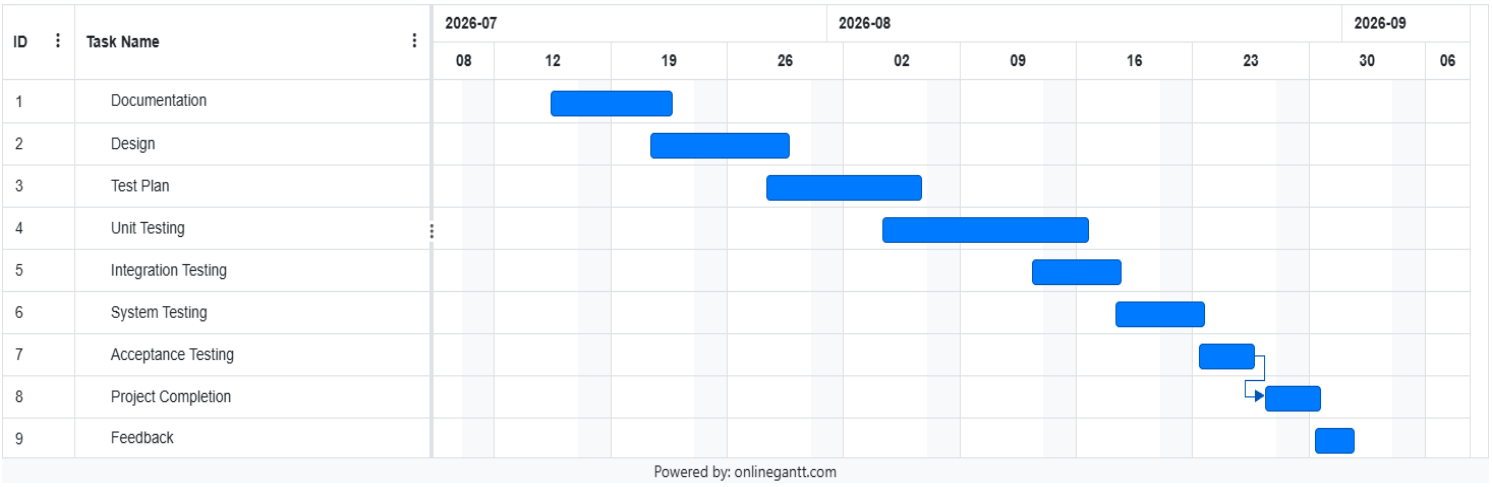
10.2 Training Needs

Training Area	Purpose
Playwright Automation Testing	Automated functional testing
Apache JMeter	Performance and load testing
OWASP ZAP	Security vulnerability testing
API Testing with Postman	Backend API validation
Git & GitHub Workflow	Version control and collaboration
Cloud Deployment	Server deployment and maintenance
Figma (UI/UX Review)	Validating built UI against approved prototypes

11. RESPONSIBILITIES

	TM	PM	Dev Team	Test Team	Client
Acceptance test Documentation & Execution	X	X		X	X
System/Integration test Documentation & Exec.	X		X	X	
Unit test documentation & execution	X		X	X	
System Design Reviews	X	X	X	X	X
Detail Design Reviews	X	X	X	X	
Test procedures and rules	X	X	X	X	
Screen & Report prototype reviews			X	X	X
Change Control and regression testing	X	X	X	X	X

12. TESTING SCHEDULE



13. PLANNING RISKS AND CONTINGENCIES

The StayEase project may face several technical, operational, and management risks. Proper contingency plans are prepared to minimize project impact.

Risk	Impact	Probability	Contingency Plan
Server Failure	System downtime; booking and search services become unavailable to Guests and Hosts	Medium	Use a backup server and enable automatic recovery system
Security Breach	Exposure of guest personal data and payment information	Medium	Apply SSL/TLS encryption and regular security testing
Internet Connectivity Issues	Service interruption	High	Use retries mechanisms and offline data caching
Payment Gateway Failure (bKash/SSL Commerz)	Guests are unable to complete booking payments, leading to abandoned bookings	Medium	Provide multiple payment gateway options and
Google Maps API Failure	Map view of room locations becomes unavailable in search results	Medium	Fall back to a text-based address display or enable manual location sharing
High User Traffic (Peak Season)	System slowdown or crash during holiday/peak booking periods	High	Use cloud auto-scaling and load balancing to support the target of 2,500+ concurrent users
Third-Party API Failure (Email/SMS)	Booking confirmations and host notification alerts are not delivered	Medium	Maintain a fallback notification channel and implement automatic retry/queueing for failed notifications
Delayed Development	Project timeline extension	Medium	Add buffer time to the schedule and conduct weekly progress monitoring

Risk	Impact	Probability	Contingency Plan
Data Loss	Loss of booking, payment, or listing records	Low	Perform daily automatic database backups
Team Member Unavailability	Development and testing delays	Medium	Cross-train team members and redistribute tasks as needed
Critical Software Bugs	System instability affecting bookings or payments	Medium	Prioritize immediate bug fixing and issue emergency patch releases
Low Host/Guest Adoption	Insufficient listings or bookings to sustain platform growth	Medium	Run local marketing campaigns and offer reduced service fees for early-adopter hosts
Fraudulent Listings or Reviews	Loss of guest trust and damage to platform reputation	Low	Require Admin approval before listings go live, and moderate reported reviews before publication